

Eli Lewis

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EDUCATION

University of Illinois at Urbana-Champaign

August 2023 – May 2027

Bachelor of Science in Aerospace Engineering — Minor in Astronomy

GPA: 4.00/4.00

Relevant Coursework: Orbital Mechanics, Aircraft Flight Mechanics, Spacecraft Rendezvous and Proximity Operations

Academic Honors: State Farm National Merit Scholarship, Engineering Visionary Scholarship, Dean's List (5 Semesters)

EXPERIENCE

Draper

Jan 2026 – June 2026

GNC Engineer Co-op

Cambridge, MA

- Developed Draper's in-house 6DOF GNC Simulink framework supporting air and space vehicle programs, implementing modular vehicle dynamics, actuator models, and sensor suites
- Formulated non-linear EKF and UKF estimation architectures, re-deriving measurement Jacobians to ingest novel sensor data and increase state observability, improving mission success rate by 43%
- Designed and validated non-linear flight control algorithms for an under-actuated, under-sensed air vehicle, improving closed-loop vehicle performance by 26%
- Architected Monte Carlo simulations to characterize performance across various mission and environmental parameters, defining system and component level requirements
- Built high-fidelity vehicle dynamics models in MATLAB and Simulink to verify physical models and define system-level GNC requirements

Machine Learning Reliability Research – Dept. of Aerospace Engineering

Jan 2025 – Dec 2025

Undergraduate Research Assistant

eli-lewis.com/checksum.html

- Built and trained neural networks using PyTorch to output a prediction reliability metric for detecting anomalous data
- Integrated Soft-Checksums with uncertainty quantification to enhance reliability in critical engineering applications
- Developed a composite loss function combining prediction accuracy with checksum penalty and reward terms
- Achieved 99.7% AUROC on a 0D combustion dataset, cutting undetected out-of-distribution samples from 97.7% to 0.1%

PROJECTS & LEADERSHIP

Liquid Rocketry at Illinois

Aug 2024 – May 2025

Controls Sub-Team Lead

Champaign, IL

- Directed an interdisciplinary GNC software team in C++ and Simulink to control roll rate for a model rocket launch
- Automated ANSYS CFD workflow with Python to model non-linear fin deflection behavior, generating aerodynamic coefficient lookup tables to parameterize vehicle roll-control authority
- Sized actuated fin deflector geometry and deflection limits ($\pm 15^\circ$), maximizing roll-control authority and maneuverability while maintaining passive longitudinal stability across the flight envelope

Reaction-Wheel Guided Baseball Simulation

July 2026 – August 2026

Personal Project

eli-lewis.com/baseball.html

- Built a 6-DOF quaternion rigid-body simulation in Python of a reaction-wheel-actuated vehicle
- Implemented an 11-state Extended Kalman Filter estimating velocity, attitude, body rate, and aerodynamic scaling factor
- Derived a closed-form guidance law mapping terminal position error to a commanded inertial spin vector
- Ran Monte Carlo sweep over release dispersion, moving median miss 0.42 m to 0.01 m and strike rate from 25% to 100%

SKILLS

Programming Languages: Python, PyTorch, C++, Linux, MATLAB, Simulink

Technical Skills: Guidance Navigation and Control (GNC), Estimation (EKF/UKF), 6DOF Dynamics, Verification & Validation, Model Based Systems Engineering (MBSE), Numerical Methods, Simulation and Modeling